

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: *Developing a Validated, Multi-Centre AI-Based Triage Decision-Support System for Emergency Departments*

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Statement of Problem

Triage nurses at Mercer General Hospital face patient prioritisation decisions under conditions of seasonal surges, staff shortages and increasing emergency department (ED) demand. Traditional triage systems such as the Canadian Triage and Acuity Scale (CTAS) and Manchester Triage Scale (MTS) rely on clinical judgement, which produces undertriage or overtriage in an estimated one in six assessments (Porto, 2024). In Caribbean Emergency Departments, these pressures are heightened by limited resources and the absence of validated decision-support tools. Existing AI-assisted triage models have demonstrated strong predictive performance but have not been validated on Caribbean patient populations. This project proposes a 12-week pilot to evaluate whether a machine-learning (ML) model trained on Mercer ED data can reduce triage level inconsistencies by $\geq 15\%$ compared to nurse-assigned categories.

Previous Work - Literature Review

Patipan Sitthiprawiat et al. (2025)

Problem: Existing triage systems such as CTAS may not consistently identify a patient's risk of deterioration, ICU admission or mortality. Patipan Sitthiprawiat et al. (2025) aimed to develop AI-based models that are capable of improving early risk assessments during triage.

Methodology: Patipan Sitthiprawiat et al. (2025) analysed 163,452 ED visits from a hospital in Thailand between 2018-2022. The study tested XGBoost, a prediction-focused ML algorithm, Logistic Regression and Random Forest against CTAS. Patient complaints were processed using a multilingual sentence embedded, a natural language processing (NLP) technique that converts written text into numerical representation for ML algorithms.

Results: XGBoost achieved the highest performance, with an Area Under the Receiver Operating Characteristic Curve (AUROC), a measure of how well a model distinguishes between outcomes, of 0.917 compared to 0.882 for CTAS and an Area Under the Precision-Recall Curve (AUPRC) of 0.629 compared to 0.333. This demonstrates the ability for the ML model to identify rare critical events. The strongest predictors were mode of arrival, patient age and patient complaints.

Limitations: The study excluded patients that were initially admitted to the general ward, who then later deteriorated and required ICU transfer, potentially underestimating the number of critically ill patients and overestimating the model's performance. The study was also done using a single institution, this limits the viability of the ML model's performance to other clinical settings.

Porto (2024)

Problem: Porto (2024) investigated which ML and NLP techniques are most effective for improving ED triage performance, given that traditional systems such as the MTS can be influenced by subjective judgment and may not consistently identify high-risk patients.

Methodology: Porto (2024) conducted a systematic review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, 60 studies were done which covered 57 ML algorithms. The review included XGBoost, Gradient Boosting which improves accuracy by combining multiple decision trees, and Deep Neural Networks, which learn complex patterns from large datasets. NLP models were also evaluated for their ability to analyse clinical information such as patient complaints and triage notes.

Results: XGBoost, Gradient Boosting and Deep Neural Networks achieved the strongest performance, with NLP techniques further improving triage classifications. The best predictors of patient outcomes were oxygen saturation, patient complaints, systolic blood pressure, age and mode of arrival.

Limitations: Many studies reviewed carried a high risk of bias due to their reliance on previous data, inconsistent reporting and limited external validation. Significant differences in datasets, algorithms and evaluation methods found in this study also made comparisons challenging.

Adam et al. (2022)

Problem: Adam et al. (2022) examined the practical dangers of deploying AI-based models in ED decision making. While ML and AI tools are used to support clinical decisions, the impact of bias on clinical behaviour, especially in the ED, was not well understood.

Methodology: Experimental methods were used to test how clinicians and non-experts respond to prescriptive vs descriptive AI recommendations. Participants were exposed to the outputs from a biased algorithm and their decisions were measured. The study specially assessed how much bias affected decision-making depending on how the AI output was framed, whether it was a direct recommendation or a descriptive flag.

Results: The biased AI recommendations affected both non-experts and clinicians, especially in situations that involved racial or religious minorities. Crucially, Adam et al. (2022) discovered that participants were able to maintain their initial decision-making when the AI output was framed as descriptive flags rather than prescriptive flags.

Limitations: The study was simulated using emergency scenarios rather than real clinical environments, which limits the finding's applicability to actual ED workflows.

Rosenbacke et al. (2024)

Problem: Rosenbacke et al. (2024) investigated a key conflict in the application of clinical AI: whether increasing the explainability of AI systems genuinely increases clinician trust and whether greater trust always results in better decisions. The study challenged the notion that explainability of AI in the healthcare system is advantageous.

Methodology: Rosenbacke et al. (2024) conducted a systematic review following PRISMA guidelines. PubMed, Web of Science and other databases for studies examining the relationship

between explainable AI (XAI) and clinician trust in healthcare applications were used. Studies included if they measured trust as an outcome in response to XAI in clinical decision-making.

Results: Most research indicated that XAI can improve clinician trust. However, Rosenbacke et al. (2024) found that overconfidence can cause clinicians to defer to AI outputs even when the output contradicts sound clinical judgement, this can then actively undermine trust, Therefore, explainability and trust did not have a linear relationship.

Limitations: Comparison and overall synthesis were difficult because the studies' definitions and evaluation of trust differed greatly. Furthermore, only a small number of studies were conducted in the ED setting.

Stelios Boulitsakis Logothetis et al. (2023)

Problem: Stelios Boulitsakis Logothetis et al. (2023) addressed key gaps in ED ML research: the majority of high-performing models function as “black boxes” by generating precise predictions without providing any explanation. Stelios Boulitsakis Logothetis et al. (2023) attempted to create a ML model for predicting acute patient deterioration, arguing that predictive accuracy alone is insufficient for safe clinical deployment

Methodology: Stelios Boulitsakis Logothetis et al. (2023) analysed patient data from electronic patient records from the point of ED entry. Three ML models were compared: logistical regression, gradient boosted decision trees and support vector machines (SVM), a ML model that separates different categories of data by drawing the widest possible boundary line between them.

Results: All three models demonstrated useful predictive performance for inhospital mortality and clinical care admission. Importantly, the study found that the models achieved competitive accuracy in comparison to the black-box alternatives.

Limitations: The study was conducted at a NHS trust in the United Kingdom, limiting the generalisability to clinical settings with different patient populations, documentation practices or data infrastructure.

Gap Analysis:

Gap 1: Lack of Caribbean Validation

Porto (2024) reviewed 60 studies covering 57 ML algorithms for ED triage and found that all validated models were developed using patient data from Asia, Europe or North America, with no study identified that developed or validated an AI triage model using Caribbean data. This gap is significant as Caribbean clinical settings frequently serve distinct patient populations, often rely on paper-based record systems and experience seasonal surges driven by visitor populations. Developing and internally validating a machine learning model on regional ED records is required to generate the first evidence of AI-assisted triage within a Caribbean clinical setting.

Gap 2: Limited Evaluation in Real-World Clinical Settings

Porto (2024) identified the use of previously collected data and the absence of workflow evaluation as a limitation across the AI triage literature, while Rosenbacke et al. (2024) demonstrated that clinical trust in AI systems is shaped by not only model accuracy but how the outputs are presented within real clinical settings. Triage nurses operate under high-pressure conditions that dataset-based studies do not capture, including strict time constraints, paper-based intakes and understaffing during peak surges. Prospective cynical evaluations that compare AI recommendations directly against nurse-assigned categories is necessary as it provides the implementation-focus workflow that the existing literature lacks.

Proposed Solution:

I propose developing and evaluating an AI-assisted triage decision support tool using historical ED data from Mercer General Hospital. Trained on routine triage variables such as vital signs, pain score and mode of arrival, that is collected within a 3 to 5 minute window, the model aims to achieve an AUROC ≥ 0.85 and reduce CTAS and MTS triage inconsistencies by 15%

compared to nurse-assigned categories while remaining operable within Mercer's time constrained environment.

The solution addresses the lack of Caribbean specific validation in existing literature. Porto (2024) highlights that AI triage models validated in high-income, electronic settings may not transfer to resource-limited Caribbean environments. Training and validating exclusively on de-identified Mercer ED records will generate the first preliminary evidence of AI-assisted triage performance in a Caribbean clinical setting.

The project also evaluated AI systems within real clinical workflow rather than focusing on solely predictive accuracy. The pilot will compare AI-generated CTAS, MTS and GCS recommendations against nurse-assigned categories to examine agreement rates and identify critical patients. To protect nurse-led judgement and mitigate algorithmic bias according to Adam et al. (2022), outputs will contain descriptive flag risks rather than predictive directives. Furthermore, model outputs will be interpretable and explainable (Stelios Boulitsakis Logothetis et al. 2023), addressing Sister Patrice Alleyne's observation that clinical adoption is Mercer's primary barrier to new tools.

Over the 12 week project period, historical data will be used to develop and test the model. Concrete outputs include a trained and validated ML model, a performance evaluated report and a documented GitHub repository.

The expected outcome is a validated AI-assisted triage model achieving an AUROC ≥ 0.85 and a $\geq 15\%$ reduction in triage-level inconsistencies at Mercer General Hospital. These findings will provide a foundation for future prospective evaluations and a potential large scale implementation across Caribbean EDs.

References

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